INSEAD The Business School for the World®

Process and Operations Management
Tutorial 4: Managing Risks

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POM Tutorials

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Problem 4: Storewars

Consider the situation by a manufacturer of playstations, SUNNY, who is facing Christmas demand. SUNNY sells the playstations through a large retail chain, TOYS-are-MINE, at a manufacturer (wholesale) price of $500 per playstation. SUNNY's unit manufacturing cost is $ 350 per playstation. TOYS-are-MINE offers the playstations for sale at its retail stores at the retail price of $800 per station. Its demand forecast for the upcoming Christmas season is for 200,000 playstations, with a standard deviation of 50,000. The retail chain has to order ahead of the season. Any playstations ordered will be delivered in time for the Christmas sales season, but no re-supply is possible during sales season.

Any playstation that will not be sold during the upcoming Christmas season will be sold after the season at a discount, for a price of $300 per station.

Problem 4: Storewars

(2a.i) What is the optimal order quantity for TOYS-are-MINE, and what are its expected profits on this item?

We follow the steps in Recipe #3. The critical ratio is

$$\frac{C_u}{C_u + C_o} = \frac{r - c}{r - s} = \frac{800 - 500}{800 - 300} = \frac{300}{500} = 0.60$$

We search for a z-value such that Φ(z)= 0.60.

Looking in Table 1 we find that the critical ratio value falls between two entries on the table z_1 = 0.25 and z_2 = 0.26.

Applying the round-up rule (closest from above), we choose the entry with the larger quantity, i.e. z^* = 0.26.

Therefore, Q^* = μ + z^* · σ = 200, 000 + 0.26 · 50, 000 = 213, 000 PlayStations.

Problem 4: Storewars

(2a.i) What is the optimal order quantity for TOYS-are-MINE, and what are its expected profits on this item?

Next, we calculate the expected profit of the retailer TOYS-are-MINE (TaM).

Use the z-value you found before to look up in the Standard Normal Loss Function Table 2 the corresponding L(z)-value.

TaM's Expected lost sales = $\sigma \cdot L(z^*) = 50,000 \cdot 0.28235 = 14,117.5$ PlayStations

TaM's Expected sales = μ − Expected lost sales = 200,000 − 14,117.5 = 185,882.5 PlayStations

TaM's Expected leftover inventory = Q^* − Expected sales = 213, 000 − 185,882.5 = 27,117.5 PlayStations

TaM's Expected profit = [(r − c) * Exp. sales] − [(c − s) * Exp. leftover inv.] = [($300) * 185,882.5] − [($200) * 27,117.5] = $50,341,250

Problem 4: Storewars

(2a.ii) What is the expected profit of the manufacturer, SUNNY?

SUNNY's profit from selling the PlayStation at TOYS-are-MINE store is 213, 000 * $150 = $31, 950, 000, where 213, 000 is the number of PlayStations that TOYS-are-MINE purchases and $150 is SUNNY's gross margin $500 − $350 = $150.

While TOYS-are-MINE might be quite pleased with this situation (it does earn $50m relative to SUNNY's $32m approximately), it should not be.

The total supply chain's profit is $31, 950, 000 + $50, 341, 250 = $82, 291, 250, but it could be higher as we explain next.

Problem 4: Storewars

(2b) What is the optimal order quantity on this item, if the retailer and the manufacturer were integrated into a single company?

Here we need to find the supply chain's optimal quantity, since this is the quantity that maximizes the integrated supply chain.

We can still use the newsvendor model to evaluate the supply chain's order quantity decision and performance measures.

Each lost sale costs the supply chain (SC) the difference between the retail price and the manufacturing cost, \$800 − \$350 = \$450; that is the SC's underage cost is C_u = \$450.

Each leftover PlayStation costs the SC the difference between the manufacturing cost and the salvage value, \$350 − \$300 = \$50; that is, the SC's overage cost is C_o = \$50.

The SC's critical ratio is

$$\frac{C_u}{C_u + C_o} = \frac{450}{450 + 50} = 0.9$$

Problem 4: Storewars

(2b) What is the optimal order quantity on this item, if the retailer and the manufacturer were integrated into a single company?

The appropriate z-statistic for that critical ratio is $z^* = 1.29$, because $\Phi(1.28) = 0.89973$ and $\Phi(1.29) = 0.90147$.

The SC's expected profit-maximizing order quantity is then

$Q^* = \mu + z^* \cdot \sigma = 200,000 + 1.29 \cdot 50,000 = 264,500$

which is higher than TOYS-are-MINE's order of 213, 000 PlayStations.

Problem 4: Storewars

(2c) Compare the expected profits generated by the integrated supply chain (which you computed in your answer to (2b)), with that of the chain formed by the two independent firms, SUNNY and TOYS-are-MINE. Which arrangement generates the highest expected profits for the supply chain? Why? What other major differences do you observe between these two inter-organizational arrangements?

The SC's performance measures can then be evaluated assuming the SC optimal order quantity, 264, 500 units.

SC's Expected lost sales = $\sigma \cdot L(1.29) = 50,000 \cdot 0.0465 = 2,325$ PlayStations

SC's Expected sales = μ − Expected lost sales = 200, 000 − 2, 325 = 197, 675 PlayStations

SC's Expected leftover inventory = Q^* − Expected sales = 264, 500 − 197, 675 = 66, 825 PlayStations

SC's Expected profit = [(r − c) * Exp. sales] − [(c − s) * Exp. leftover inv.] = [($450) * 197, 675] − [($50) * 66, 825] = $85, 612, 500

Problem 4: Storewars

(2c) Compare the expected profits generated by the integrated supply chain (which you computed in your answer to (2b)), with that of the chain formed by the two independent firms, SUNNY and TOYS-are-MINE. Which arrangement generates the highest expected profits for the supply chain? Why? What other major differences do you observe between these two inter-organizational arrangements?

Thus, while SUNNY and TOYS-are-MINE currently earn an expected profit of $82,291,250, their SC could enjoy an expected profit that is about 4 percent higher, or $3,321,250.

But, why the current supply chain perform worse than it could? The obvious answer is that TOYS-are-MINE does not order enough PlayStations: TOYS-are-MINE orders 213, 000 of them, but the SC's optimal order quantity is 264, 500 units.

But why doesn't TOYS-are-MINE order enough? Because TOYS-are-MINE is acting in its own self interest to maximize its own profit.

Problem 4: Storewars

(2c) Compare the expected profits generated by the integrated supply chain (which you computed in your answer to (2b)), with that of the chain formed by the two independent firms, SUNNY and TOYS-are-MINE. Which arrangement generates the highest expected profits for the supply chain? Why? What other major differences do you observe between these two inter-organizational arrangements?

To explain further, TOYS-are-MINE must pay SUNNY $500 per PlayStation and so TOYS-are-MINE acts as if the cost to produce each PlayStation is $500, not the actual $350.

From TOYS-are-MINE's perspective, it does not matter if the actual manufacturing cost is $350, $400, or even $0; its "manufacturing cost" is $500.

This is related to an important phenomenon called double marginalization. In simple words, TOYS-are-MINE only earns a "piece of the pie" ($300) of the total "pie" benefit per sale ($450). Thus, it is logical that TOYS-are-MINE is not willing to purchase as much inventory as would be optimal for the supply chain.

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Problem 1: Banking Crisis

You are in the unfortunate but well-paid position of being Chief Investment Officer at a leading hedge fund. While your fund has been performing well, the contagion and fear in capital markets is getting to your investors. This is leading to huge redemptions from your fund. You must plan to meet these redemptions at the lowest costs. You and your economic team employ the Quaker method of estimating these redemptions and predict that over the course of the next month, the expected redemptions will be US$ 300 Million. The standard deviation of this estimate is US$100 million. Unfortunately, all your hedge fund assets are illiquid and cannot be disposed for the next year. You must thus go to the capital markets for raising additional funds.

Problem 1: Banking Crisis

You have two sources of financing that you are planning to use simultaneously:

Source 1: You can borrow from the government sponsored TARP (The-Augmentation-of-Rich-People) program. This facility charges a usurious 25% annual rate of interest. Further, it can be availed only if a request is made far in advance.

Source 2: Alternately, when TARP funding comes short, you can borrow from the overnight credit markets. You can use this facility anytime and it provides funding immediately. Unfortunately, the rate of interest charged is 40% per annum.

Newsvendor setup:

What is my quantity decision in the presence of demand uncertainty?

What do I lose if I have too small or too large quantity?

Problem 1: Banking Crisis

a) How much money would you borrow from the TARP funds?

This is a classic newsvendor setup with reactive capacity: The cost of borrowing too little is $C_u = 15$, whereas the cost of borrowing too much is $C_o = 25$.

Critical ratio is 15/(15+25) = 3/8=0.375. From the table, z-value is -0.31 (remember, if we can't find the exact value, we look for the next highest value).

So the amount borrowed is

$Q^* = \mu + z^* \times \sigma = 300 - 0.31 \times 100 =$ US$ 269 million.

Problem 1: Banking Crisis

b) What are your annualized capital costs if you employ the solution prescribed in part A?

For 269 million you have to pay 25%, that is 269*0.25 = 67.25 million.

On the other hand, because you need to satisfy whatever demand there is, you will have to borrow from the overnight market if there is demand and you don't have the money available. In other words, if anything above 269 million is required, you have to borrow.

This means that you have to borrow the expected lost sales from the overnight market.

For z* of -0.31, from the table you read: L(z*) = 0.57296.

Expected lost sales are then: 100*0.57296 = 57.30 million.

For this amount you need to pay 40%, or 22.92 million.

Total costs 67.25 + 22.92 = 90.17 million.

Problem 1: Banking Crisis

c) One of your classmates at INSEAD is in exactly the same financial position as you. While he runs an identical fund, most of his clients are located in a different geography. He calls you up and proposes merging your two funds. He somewhat remembers the POM class and suggests that merging would reduce the above mentioned capital costs for both funds. Do you agree? Justify your answer in no more than 100 words.

Yes, merging the two funds will decrease capital costs for both as you will now aggregate demand for redemptions from both markets.

This will have lower uncertainty and this will lead to the capital costs being lower.

This is the same principle of aggregating demand from multiple markets that we have seen multiple times in class.

Problem 1: Banking Crisis

d) The crisis is getting worse. The overnight credit markets have now completely frozen up. You no longer care about minimizing your capital costs. All you care about is the odds of your survival, or the probability that your fund will not be able to meet redemptions. What is the minimum amount of money you need to borrow from TARP funds to ensure that you will be able to meet all redemption requests with a 95% probability?

This translates to a Q such that Pr(D <= Q) = 0.95.

We can rewrite this as

Pr ((D-300)/100 <= (Q-300)/100) = 0.95.

Because demand is normal with mean 300 and standard deviation 100, then (D-300)/100 is standard normal.

So the above probability is equal to: Pr(Z <= z) = 0.95 with z = (Q-300)/100. From the table the value of z corresponding to 0.95 is 1.65 and therefore: z = Q-300/100 = 1.65 or Q = 465 million.

Problem 2: Energy Economics

A massive natural gas field has been discovered in the beautiful Central Asian Republic of Gasistan. The gas field provides a limitless supply of gas that has the potential to transform the lives of the people (and rulers) of Gasistan. However, to realize this prosperity the National Gas Corporation of Gasistan (NGCG) must develop facilities to transport the plentiful natural gas to energy-hungry markets. The geological makeup of this field dictates that gas that is produced must be transported to markets or it is lost. Assume that for the time-frame of the analysis, the field can produce any quantity of gas that can be transported. You have been hired as a consultant to the NGCG.

Based on your research, you have outlined two potential ways to transport gas to somewhat differing markets.

Problem 2: Energy Economics

1. Pipeline: A pipeline can be built to transport a fixed quantity of gas to European markets. Daily demand in the European markets that can be accessed by the proposed pipeline is distributed Normally with a mean of 600 Million Cubic Feet (mcf) and a standard deviation of 600 mcf. The low operation costs of the pipeline ensure that every mcf of gas that can be sold in European markets, NGCG earns a profit of $10,000 (taking into account <u>operating costs</u>, i.e., production, transportation and all other operating costs). Building a pipeline involves a heavy capital investment. Based on the availability of capital and the sovereign guarantees available to the NGCG, you estimate that the daily <u>capital cost</u> of one mcf of transportation capacity will be $5,000. In other words, if a pipeline with daily capacity of 1,000 mcf is built, the daily capital cost would be $5 million.

Problem 2: Energy Economics

2. Liquefying and shipping terminal: Natural gas may also be liquefied and can then be sent via special sea-going tankers to a variety of global markets. Daily Demand in the markets that can economically be accessed by sea-going vessels is distributed normally with the same mean as daily demand in European markets, 600mcf. However, the diversity of markets that can be accessed by sea-going vessels limits the impacts of region-specific demand shocks and the standard deviation of the accessible markets is only 200 mcf. Further, you estimate that every mcf of gas that can be sold in these markets will earn a profit of $9,000. Finally, you estimate that the daily capital cost of one mcf of liquefying and transportation capacity will be $6,000.

Problem 2: Energy Economics

a) If you decide to build a pipeline, what is the capacity of the pipeline that maximizes the expected profits?

NCGC is facing uncertain demand from the market and has to make a bet on this potential demand, by installing pipeline capacity. A huge pipeline is likely to go unused while with a small pipeline NCGC is likely to lose out on potential sales. The newsvendor model applies in this setting.

To obtain the optimal capacity, we need to calculate the underage and overage costs.

C_u = 10K – 5K = 5K.

10K is the revenue and 5K is the cost of installing one unit.

C_o = 5K.

Thus the critical factor is $C_u/(C_u+C_o) = 0.5$.

This means that in this case NCGC needs to build a capacity of 600mcf. (Note that $z^* = 0$).

Problem 2: Energy Economics

b) If you decide to build a pipeline with the capacity calculated in A., what would be the average utilization of that pipeline?

Utilization is the fraction of the installed capacity gets used (on average). To compute this number, we need the installed capacity and the average capacity used. We already know the total capacity from A. We need to next compute the average capacity used.

Average capacity used corresponds to the average sales in a standard newsvendor setup. So we need to compute the expected sales:

E[Sales] = E[Demand] – E[Lost Sales] = 600-600*L(0) = 600-600*0.39894 = 360.64.

Utilization = 360.64 / 600 = 60.11%.

Problem 2: Energy Economics

c) With the pipeline option, what are the daily profits? (taking into account the capital costs)

Profits = Expected sales*10K - Capacity*5K = 360.64*10,000 - 600*5,000 = $606,360.

Problem 2: Energy Economics

d) If you decide to build the liquefying facility, what is the profit maximizing capacity of the liquefying and shipping terminal?

We need to repeat the same procedure as for pipeline option.

It is easy to see that $C_u = 3000$ and $C_o = 6000$ and thus the critical fractile is: 1/3. From Table 1, we have: $z^* = -0.43$ (note this is negative).

Therefore: $Q^* = \mu + z^* \times \sigma = 600 - 0.43 \times 200 = 514$mcf

Problem 2: Energy Economics

e) If you decide to build the liquefying facility with the capacity calculated in D, what would be the utilization of that facility?

E [Sales] = E [Demand] - E[Lost Sales] = 600 - 200*L(-0.43) = 600 - 200*0.65027 = 469.95.

The Utilization = 469.95 / 514 = 91.43%.

Problem 2: Energy Economics

f) With the liquefying option, what are the daily profits? (taking into account the capital costs)

Profits = Expected sales*9K - Capacity*6K = 469.95*9,000 - 514*6,000 = $1,145,514.

Problem 2: Energy Economics

g) Based on your computation of daily profits, which of the two options would you chose. Qualitatively, highlight the key difference between the two options that drives your recommendation.

Go for the Liquefying option, because liquefying allows for aggregation of demand from multiple markets. Demand aggregated from multiple markets has lower variance than demand from one market.

If demand variance (or uncertainty) is lower, firm profits are higher. Here we see that this can be the case even if the per-unit profit is substantially lower.

Problem 3: Land's End

Geoff Gullo owns a small firm that manufactures "Gullo Sunglasses". He has the opportunity to sell a particular seasonal model to Land's End. Geoff offers Land's End two purchasing options:

Option 1: Geoff offers to set his price at $65 and agrees to credit Land's End $53 for each unit Land's End returns to Geoff at the end of the season (because these units did not sell). Since styles change each year, there is essentially no value in the returned merchandise.

Option 2: Geoff offers a price of $55 for each unit, but returns are no longer accepted. In this case, Land's End throws out unsold units at the end of the season.

The season's demand for this model is Normal with mean 200 and standard deviation 125. Land's end will sell those sunglasses for $100 each. Geoff's production cost is $25.

Problem 3: Land's End

a) How much would Land's End buy if they chose Option 1?

With option 1 Land's End sales price is \$100, purchase cost is \$65 and salvage value is \$53 (because Geoff buys back unsold glasses for \$53).

So the underage cost is C_u =100 – 65 = 35 and the overage cost is C_o = 65 – 53 =12. The critical ratio is $C_u/(C_o + C_u)$ = 35/47 = 0.7447. From the Standard Normal Distribution Function Table we choose z = 0.66.

The optimal order quantity is then $Q = \mu + \sigma * z = 200 + 0.66*125 = 283$.

Problem 3: Land's End

b) How much would Land's End buy if they chose Option 2?

With option 2 Land's End sales price is \$100, purchase cost is \$55 and the salvage value is \$0.

So the underage cost is $C_u = 100 - 55 = 45$ and the overage cost is $C_o = 55$. The critical ratio is $C_u/(C_o + C_u) = 45/100 = 0.4500$.

From the Standard Normal Distribution Function Table choose $z = -0.12$. The optimal order quantity is then $Q = \mu + \sigma * z = 200 - 0.12*125 = 185$.

Problem 3: Land's End

c) Which option will Land's End choose?

We need to evaluate expected profits in each case to determine which option Lands End should choose. The process to calculate the expected profits is mechanical, as follows:

Expected lost sales: 125*L(0.66) = 125*0.15281=19.10

Expected sales: mean demand - expected lost sales = 200 – 19.10 = 180.90

Expected leftover inventory = Q* - expected sales= 283 – 180.90 = 102.10

Expected profit = C_u * expected sales - C_o * expected leftover inventory = 35*180.90 – 12*102.10 = 5106.30

We now need to repeat the same process for option 2. This results is an expected profit of 4052.38. Therefore, option 1 should be chosen.

Problem 3: Land's End

d) Suppose Land's End chooses option 1 and orders 275 units. What is Geoff Gullo's expected profit?

REMARK 1: This question asks for GEOFF GULLO's profits and not Land's End profits.

REMARK 2: This question says that Land's End orders 275 units and not the optimal order quantity calculated before. All the calculations must be based on the new order quantity and not the optimal one.

Problem 3: Land's End

d) Suppose Land's End chooses option 1 and orders 275 units. What is Geoff Gullo's expected profit?

Geoff first needs to produce 275 which cost 25 each. Then, it sells them to Land's End at 65 each. But it also needs to take back the unsold ones (or expected leftover inventory for the Land's End).

The only component we don't have is the expected leftover inventory, which we calculate as follows:

First find the z-value corresponding to new order quantity: z = (275 – 200) / 125 = 0.6.

Then we have:

Expected lost sales = 125 * L(z)= 125 * 0.16867 = 21.09

Expected sales = 200 – 21.09 = 178.92

Expected leftover inventory = 275 – 178.91 = 96.08

Therefore:

Geoff's Profit: 275*65 – 275*25 – 96.09*53 = 5907.56